

# UX Portfolio: Understanding User Experience

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## 1.0 Introduction

User Experience (UX) is a field dedicated to studying, designing, and evaluating users' interactions with systems, products, or services. The resulting experience is influenced by the user's context, environment, and the user themselves. These elements contribute to the creation of a narrative; a memorized experience. Design looks to capture this narrative using various methods, models, and approaches to translate it into a design that not only meets user needs but also aligns with their values, fostering meaningful interactions between the user and the system, product, or service.

This portfolio aims to deepen UX understanding by providing a comprehensive view, summarizing methods, models, and approaches from lectures and self-study for future research and implementation. My learning goals are to understand UX and explore methods for future projects.

## 2.0 UX Definitions and Descriptions

The multidisciplinary nature of UX yields diverse interpretations from psychology to business, from quality-centric to value-centric perspectives (1). UX is seen as a phenomenon, a field of study and practice, and a method for crafting meaningful products.

### 2.1 User experience as a phenomenon

UX covers interactions of a user with systems, including both active and passive engagements. These interactions are influenced by prior encounters and shaped by social/cultural contexts (1).

This definition emphasizes moment-to-moment actions, with the experience varying based on the time span involved. Anticipation UX involves users' expectations, while Momentary UX occurs during interaction, Episodic UX involves recollection afterward, and Cumulative UX reflects the impact of multiple encounters over time (1).

Viewing UX as a phenomenon raises questions about how we understand, study, and evaluate

user experiences. This perspective prompts the question: Do designers design experiences, or do they design for experiences?

## 2.2 User experience as a field of study and practice

The emergence of UX as a field of study arose from questioning the importance of Usability versus the broader concept of Experience, putting at the center of the process, the user's needs (7). Designers then developed new methods that considered emotional, sensory, and qualitative aspects alongside functionality and usability (3,4,7).

Don Norman emphasizes the interconnectedness of products and experiences, highlighting the need to design the entire process seamlessly, from acquisition to maintenance (26). UX designers prioritize aspects like pleasure, efficiency, and fun, ensuring specific user needs are met within the product interaction context (2). Principles of human-centered design guide UX, emphasizing user centrality, identification of crucial aspects, and iterative design involving user participation (1).

UX design diverges from traditional human-centered design by incorporating several factors such as methods, tools, criteria, and organizational positioning, significantly impacting design suitability and usage in diverse situations (1).

## 2.3 User experience to create meaningful products

Considering the quote, "experience or user experience is not about technology, industrial design, or interfaces. It is about crafting meaningful experiences through a device" (2), UX dives into understanding these narratives and crafting meaningful experiences. This involves identifying crucial elements, determining them to be personal or shared in nature, and translating them into designs.

UX, seen as a sub-category of Experience, focuses on interactive products. The challenge lies in intentionally creating and shaping experiences (2). Additionally, "people create meaning with other people, and it is this meaning that drives the uses of design and technology" (3). This approach involves understanding user narratives and meanings associated with products and environments. Design empathy fosters reciprocal relationships between design and users throughout the design cycle (3).

UX is diverse, starting with individual interactions observed and understood through various analytical methods and tools. Insights gathered inform the creation of meaningful product experiences tailored to users.

## 3.0 Reflection on weekly logbook / activities:

I chose this course to explore designing learning experiences and motivating lifelong learning. My goal is to create interactive systems that inspire and motivate individuals on their learning journeys. While I have learned practical UX design methods like personas

and flows, I found them too rigid, reducing users to customers. Thus, I seek a deeper understanding of UX to co-create meaningful products with users, exploring diverse approaches for development.

The initial lecture showed UX's multifaceted layers. The speed date workshop expanded my perspective by exploring diverse opinions on UX. One classmate had an interesting comment: I want to be able to name the methods that I intuitively already use in my process. This shaped my learning goals to form informed opinions on UX and compile methods, models, and approaches for future use.

In the first week, I focused on refining my understanding of UX, connecting different points of view. Afterwards, I delved into self-study, using a snowballing approach to explore intriguing references (1,2,3,4). I compiled notes from relevant literature to create cheat sheets for future reference, including tools and approaches.

Among the researched material, Marc Hassenzahl's vision on user experience and experience design (2) resonated with me the most. The notion that UX is a means of narrating a story through a product deeply appealed to me. I connected this concept with the methods of co-constructing stories (8) and Cultural Probes (12, 13, 14) to capture firsthand experiences that could inspire the design process.

In week two, we focused on Empathy Design. This approach intrigued me, looking into first-person perspective using second-person data. While the workshop offered insights into through a three-step activity, I sought deeper

understanding through papers by Smeenk W. et al., which explained implementing empathy design in the design process (15,16,17). This challenged my belief that firsthand experiences could lead to incorrect assumptions, showing how empathy and a first-person perspective can drive innovative solutions.

The third week afforded a holistic view of the components shaping a user’s experience. My primary interest lies in behavioral understanding (5, 21), aligning with my curiosity about understanding needs and values as the roots of human experience.

However, what significantly shaped my understanding and provided the overarching theme for my learning journey was the insight into economic paradigms (6, 7, 25). Analyzing available literature, I observed the dynamic shifts in user needs and values, thereby shaping the evolution of the UX field.

4.0 Understanding UX

4.1 UX a layered concept.

4.1.1 Understanding the needs and values of the user

Our experiences are fundamentally rooted in our needs, driving our behaviors (5). The quality of an experience is closely tied to the fulfillment of universal psychological needs (2), which both guide our actions and shape our values (5).

Kerlin’s research (5) delves into the psychological needs of youth in the digital age and presents a framework for human values

during this period. It highlights the universal nature of these values, which evolve over life stages due to changing needs and external factors. Values significantly influence our daily behaviors, thoughts, attitudes, and overall experience and understanding this is crucial for effective design of user experiences (5).

4.1.2 Development of UX through the economic paradigms.

Kerlin’s research (5) indicates that needs and

values are influenced by context, particularly the evolving economic landscape that forms societal needs and values. In the work of Brand and Rocchi (6), four paradigms illustrate the evolution of user needs and values together with the creation of economic value (Figure 1).

The experiential economy emerged when industries shifted from object-centered thinking to considering the user and their experience (7). Despite this shift, the user was seen as a customer, and value was attributed to the brand, emphasizing the pre-purchase,

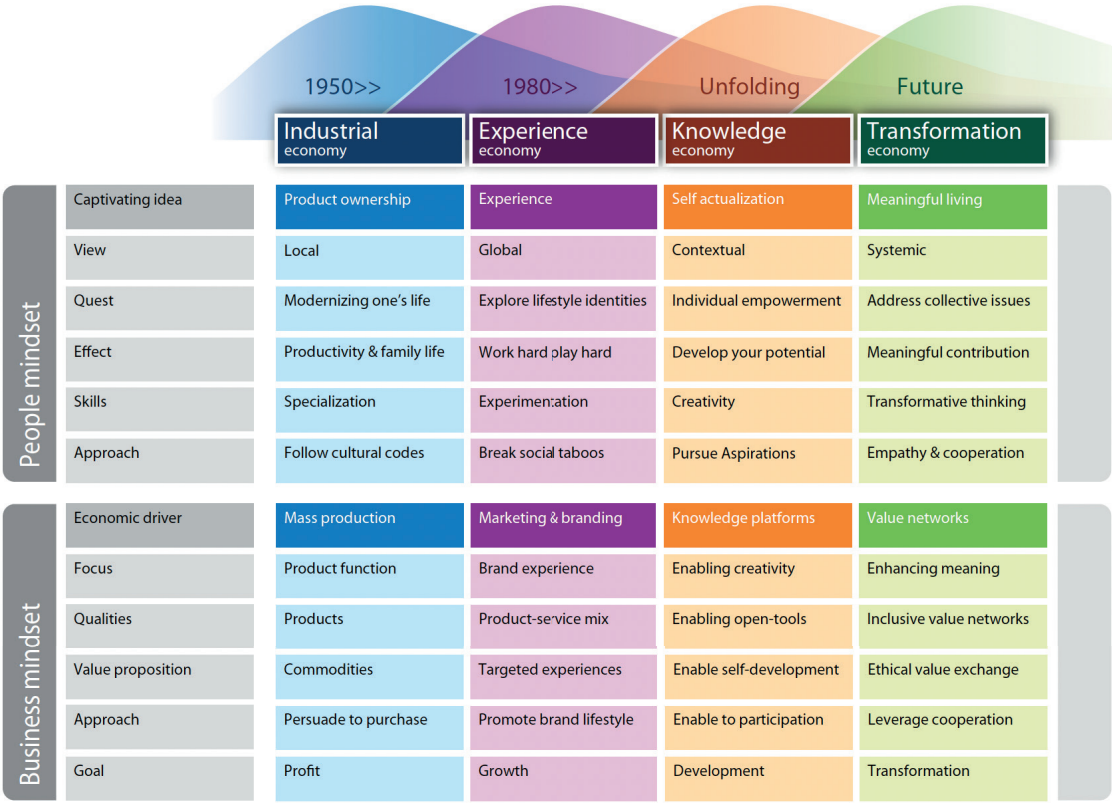


Figure 1 : Four paradigms (6)

purchase, and post-purchase experiences (7). During this period, UX was used to enhance product branding. Methods such as user observation in context, personas (Figure 2), scenarios, "total touchpoint," and experience flow (Figure 3) were developed.

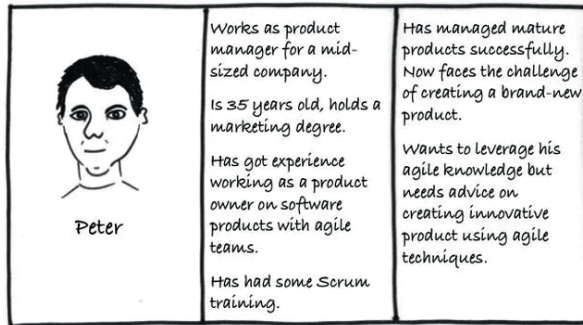


Figure 2 : Persona (27)

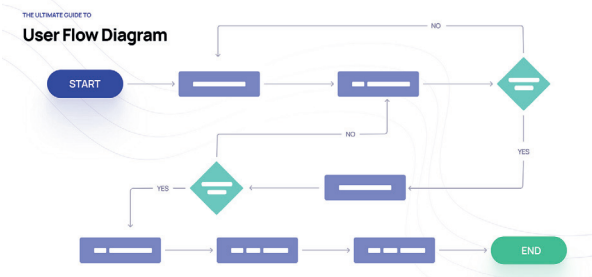


Figure 3 : Experience flow (28)

With society transitioning into an era of knowledge and knowledge sharing (6), users and their experiences became important in the design process. A robust user-centered design process emerged, incorporating users directly into the design process (7).

The availability of knowledge to designers played a pivotal role in creating value in this paradigm. Users and experts became collaborators in the design process, leading to development of methods like open-design, co-creation, participatory design, and data collection through sensors.

In the knowledge economy paradigm, the development of products based on user needs takes center stage. Providing users with positive product experiences involves satisfying their needs (2). Hassenzahl (2) suggests that asking "Why?" (Figure 4) and understanding the meaningfulness behind a product can uncover the needs and values it aims to address, creating value.

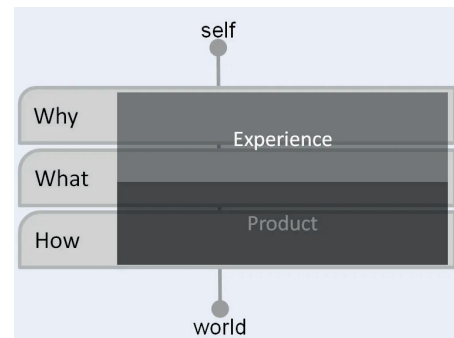


Figure 4 : The Why to the What and the How: Three levels to consider when designing technology-mediated experiences (2)

I believe that this paradigm has transformed UX from a mere design phenomenon to an in-depth study of human needs and values, aiming to create tailored and meaningful experiences for users.

#### 4.1.3 Towards UX to create meaningful systems

Hassenzahl defines experience as a memorized, personally encountered event, distinct from moment-by-moment experiences, forming memorized stories of consumption (2). The designer's role is to comprehend this mixture of feelings, thoughts, and actions.

UX can be viewed as a field trying to capture these experiences, questioning importance, uniqueness, and transferability. UX endeavors to translate these narratives into interactive products as mediators of these narratives (2).

Noteworthy methods include Co-constructive stories (8) (Figure 5), Repertory Grid Technique (RGT) (11), and Cultural Probes (12, 13, 14). Cultural Probes stand out, offering an inclusive means of gathering knowledge about individual experiences and uncovering underlying needs or behaviors through observing participants' reactions to proposed activities.

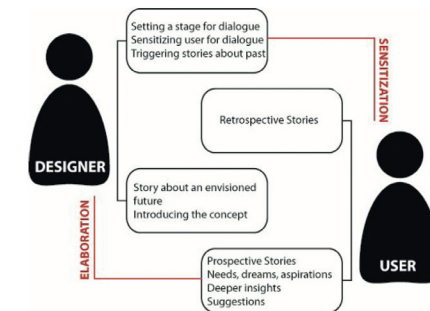


Figure 5 : Phases of co-constructing stories. (8)

However, as the knowledge economy develops and we witness the transformation of economies (6), UX becomes more



complex. User needs now cross societal and environmental considerations (6).

Addressing significant collective issues needs intimate, empathic knowledge of local contexts, needs, and cultures, achievable neither by a single stakeholder nor without such insights (7).

## 4.2 Empathy Design and UX

### 4.2.1 Empathy Design a Fundamental Aspect for UX

Koskinen's approach (3) to user experience delves deep into narratives and meanings users attach to products and environments. Design empathy is integral to the interpretative process, fostering a mutual relationship between designers and users throughout the design life cycle (3).

Design empathy refers to the ability to comprehend and share the feelings and experiences of users; essential for design interpretation, enabling designers to perceive the world from the user's viewpoint and understand their needs.

In Zaki's article (18), he suggests that empathy isn't always automatic; instead, individuals are motivated to either engage with or avoid empathic experiences. These motivations shape how we emotionally understand and connect with others. The article highlights the interplay between automatic empathic responses and the influence of specific situations on empathy.

The notion that empathy is not just an automatic reaction but can also be controlled within the right context implies that designers

can cultivate this skill to better comprehend and establish a connection with users.

### 4.2.2 Methods for empathic design.

Smeenk W. et al. (2016) introduce three different design perspectives:

- Third-person: for people and society.
- Second-person: together with a group.
- First-person: for oneself within society.

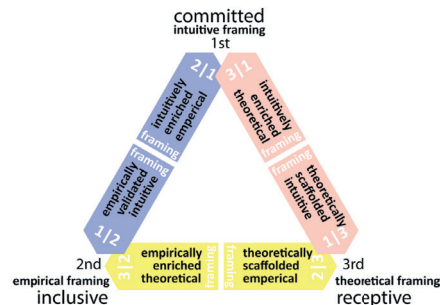


Figure 6 : Prespective transition values (16)

Each highlights different relationships between designer and user, offering a framework for understanding the design process (16). It supports a mixed perspective approach (Figure 6), promoting empathic design and enables designers to incorporate personal experiences and intuition.

Smeenk W. et al. (2018) propose an empathic approach to understanding the experience and perception of people with dementia, a fragile target group, with emphasis on safeguarding the health of participants (15). Based on three

stages (harvest, handover, ideation (Figure 7)), this approach provides novel ways to gather user data through interviews and construction scenarios based on real user experience for teammates to experience firsthand. Based on real user experiences, this technique poses as an interesting approach to empathic design.



Figure 7 : Empathic Handover Approach (15)

Furthermore, Smeenk W. et al. (2019) address the lack of a systematic overview of key dimensions and elements fostering empathy in design. They introduce an Empathic Formation Compass, defining empathic formation as the formative process of becoming an empathic design professional with the right attitude, skills, and knowledge applicable in co-design (17). The Empathic Formation Compass provides designers vocabulary to understand the dimensions and elements influencing empathic formation, informing their role and design decisions. It aims to support reflection and evaluation of co-design projects beyond method reliance, making empathic design a conscious activity where designers regulate their feelings and experiences to decrease empathic bias (17).

Although this compass has not been validated, these papers represent a positive step towards an empathic future for design.

#### 4.2.3 Empathy Design a way to radical innovation?

As our economy transitions into a transformation economy, there is a demand for significant change and innovation.

Norman's and Vergenti's paper (20) on incremental and radical innovation explores two perspectives. The first considers human/user-centered design (HCD/UCD) as methods for incremental innovation. However, it acknowledges that achieving radical innovation requires technological and meaningful change (20).

The second perspective assesses innovations through the dimensions of technology and meaning change, linking them to drivers such as technology, design, and user preferences.

Leonard's article (19) argues that empathic design is a solution to avoid stagnation in today's paradigm. This approach enables researchers to access information not available through traditional market research methods, allowing companies to uncover and address unnoticed customer needs.

Users often struggle to articulate meaning for things they have not experienced. Therefore, empathic design provides an interesting approach to stepping into the shoes of the user and identifying those hidden stories that can drive radical innovation, particularly in meaning-driven innovation (20).

Moreover, empathic design also facilitates understanding of complex human experiences.

### 4.3 Holistic approach to UX

We experience as a whole; our experience is affected by multiple factors:

#### 4.3.1 Complexity of the human experience.

"Human behavior is the expression of underlying motivations, drivers, and needs" (5), just as explicit observable actions represent human needs and values (5).

Understanding need-driven behavior requires comprehending how people experience something and is integral to understanding UX and the complexity of human experiences, helping create effective interventions (21).

However, behavior is influenced by context (5, 21) and shaped by our attitudes toward social expectations and the concept of perceived behavioral control (21).

Moreover, experience is also influenced by our level of attention. People have three levels of attention: focused, peripheral, and implicit (22). Activities frequently transition between the center and the periphery of attention in everyday life. Understanding these shifts is important in designing interactions for products in daily life.

Also, "people act on meanings created and modified in interaction with others" (3). Thus, social awareness is also crucial when designing experiences as social cues guide our decisions and behaviors. Understanding this is essential for designing interactions for UX (23).

Our decision-making process influences the way we perceive outcomes, affecting our final experience. For example, consider a "maximizer" and a "satisficer" (24). They both approach options differently. Maximizers

seek to maximize options, often leading to feelings of regret and depression. In contrast, satisficers seek a good enough option, resulting in a more positive experience.

#### 4.3.2 Models and Approaches

The Integrated Behavioral Model (IBM) (21) (Figure 8) serves as theoretical framework for understanding behavior, identifying target beliefs, and guiding interventions. Elicitation processes are critical for ensuring the relevancy and accuracy of identified beliefs, thus contributing to the success of behavioral change interventions.

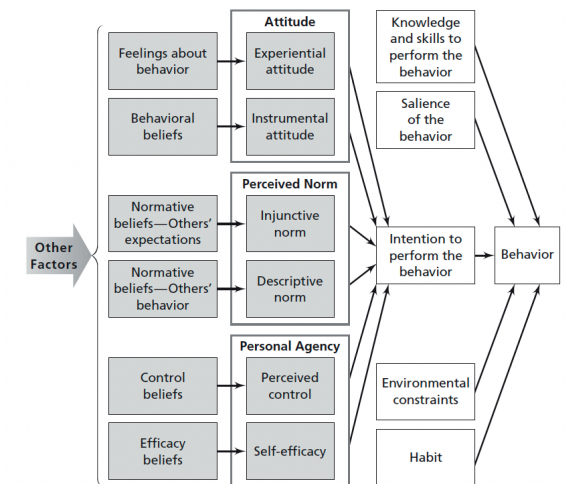


Figure 8 : The Integrated Behavioral Model (IBM) (21)

Additionally, an interesting framework to consider when designing interactions on various levels is the "interaction-attention continuum" (22). This conceptual framework aims to enhance human-centered interaction

(HCI) by accommodating various levels of human attention.

#### 4.3.3 More than human-experience

As we transition towards the transformation economic paradigm, we need to contemplate the future of user experience design.

In our journey towards the future, we are moving away from solely adopting a user-centered perspective. Instead, we are embracing a more holistic approach to experience, one that encompasses the needs and values of both humans and the environment (7). Brand further explores this concept in his article (25), delving deeper into the transformation economy and introducing a framework for contemplating co-emerging futures.

Given this shift, I find myself pondering what the realm of more-than-human experience design entails.

## 5.0 Integration of new insights in earlier work.

I would like to reflect on my graduation project, which envisioned a future where rising sea levels compelled us to coexist with water rather than resist it. The project speculated on how the landscape would appear and how society would adapt.

To illustrate this speculative concept, I created an interactive installation designed to provoke debate and curiosity. My aim was to inspire change in the way people discussed the future of our environment.



Figure 9 : Sand interaction

The project featured an engaging interaction with moving sand (Figure 9, 10), a material intuitively associated with playful experiences. The shifting sand created various landscape simulations intended to ignite curiosity.

However, I became too focused on conveying my message rather than fully exploring how the installation could tell a story.

This course helped me recognize missed opportunities from which to learn. Initially, I only considered the first- and third-person perspectives, incorporating solely my vision of the future and that of experts (16). However, I should have also considered the second-person perspective through interviews, co-constructive story sessions, and co-designing sessions. Perhaps using Cultural Probes (12,13,14) to involve participants in imagining a water-filled future would have enriched the experience by highlighting diverse viewpoints.

Furthermore, delving deeper into observing participants could have provided insights into their behavior regarding discussions about future change. Understanding what sparked interest and what provoked fear could have enabled me to craft a more tailored intervention conducive to stimulating meaningful debates for change.

Lastly, concerning the interaction design of the project, I only considered myself as a participant when testing out the mechanics. With more time, I should have organized user testing and utilized AttrackDiff to gather insights into how others perceived my installation. This would have facilitated the creation of a more engaging interaction for all.

Overall, this course has made me realize that during this project, I overlooked user



Figure 10 : Sand interaction



experience methods, models, and approaches as tools for shaping the story and conveying it through design. Instead, I focused on imposing my own perspective onto the interactive installation, hoping it would elicit a positive experience.

## 6.0 UX Proposition

I propose implementing UX in the development of alternative educational models to address current challenges. For example, UX can help reignite excitement about teaching by researching the teaching experience to reinvigorate teachers' motivation. Additionally, it can aid in bridging the social gap in education by finding ways to support students from low-income families who lack access to tutors. These are topics I'm passionate about exploring through UX to discover meaningful solutions.

Furthermore, UX offers intriguing approaches for making learning and development experiences more engaging. For instance, designing training sessions within companies to foster a mindset of lifelong learning is an area I find compelling to consider.

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